

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 8

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Recap of Previous Lecture

Forward Gaussian Diffusion Process

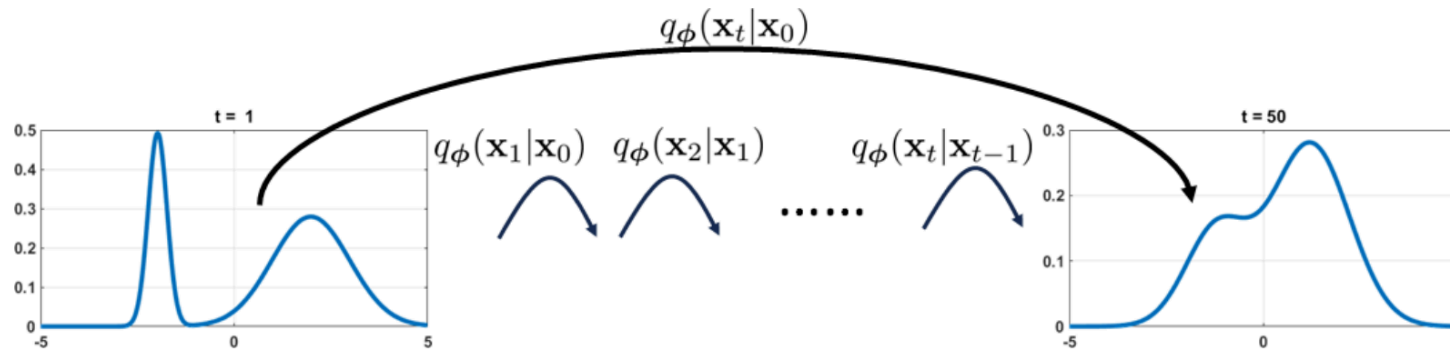
Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$, $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t, \quad \text{where } \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I});$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

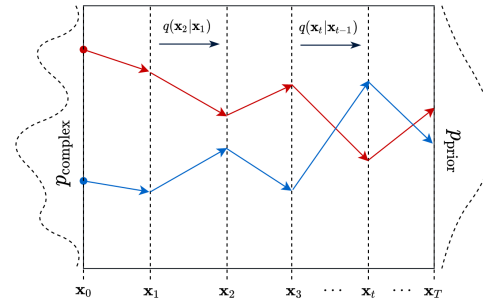
$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I});$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbf{I}).$$



Recap of Previous Lecture

Diffusion describes the migration of particles from regions of high density to those of low density.



1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}), t \geq 1$
3. After $T \gg 1$ steps: $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

If this process can be reversed, we can sample from $p_{\text{data}}(\mathbf{x})$ by starting from noise $p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$. Our goal now becomes inverting this diffusion.

Recap of Previous Lecture

NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

Gaussian Diffusion

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{1 - \bar{\alpha}_t}$$

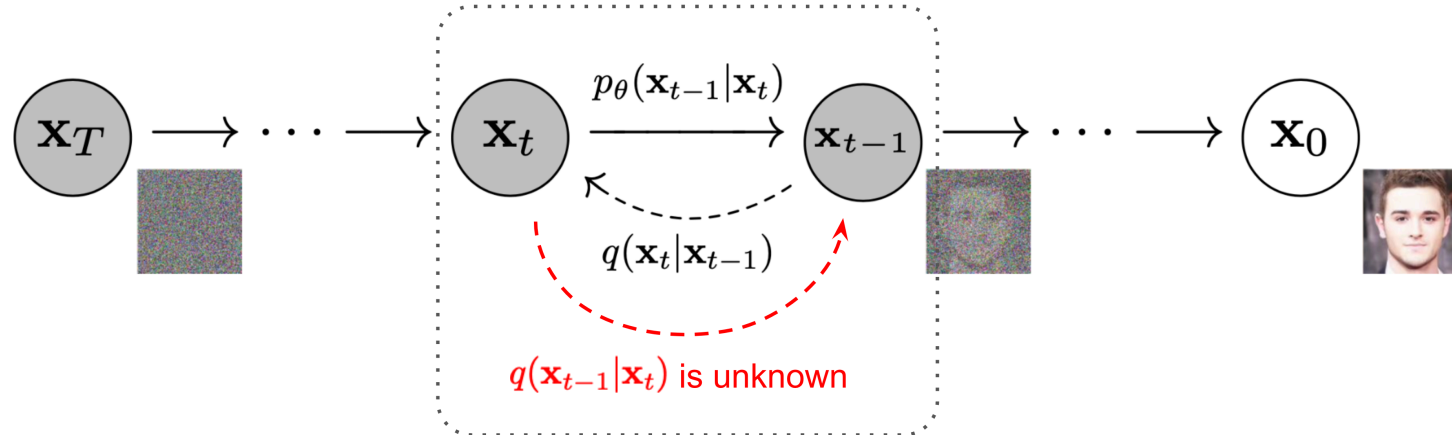
Recap of Previous Lecture

Theorem (Denoising Score Matching)

$$\begin{aligned} \mathbb{E}_{q(\mathbf{x}_t)} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)\|_2^2 \\ = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta}) \end{aligned}$$

Note: Annealed Langevin dynamics applies to diffusion, too.

Recap of Previous Lecture



Reverse Process (Ancestral Sampling)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)}$$

$$\approx p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta,t}(\mathbf{x}_t), \sigma_{\theta,t}^2(\mathbf{x}_t))$$

Feller's theorem justifies this Gaussian assumption.

Recap of Previous Lecture

Forward Process

1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \epsilon_t$
3. $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

Reverse Process

1. $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$
2. $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$
3. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$

Recap of Previous Lecture

Forward process maps any distribution $p_{\text{data}}(\mathbf{x})$ to $\mathcal{N}(0, \mathbf{I})$ by injection of noise:

$$\begin{aligned} q(\mathbf{x}_t | \mathbf{x}_{t-1}) &= \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}); \\ q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \end{aligned}$$

Reverse process refers to an intractable distribution that can be approximated by a normal distribution (with unknown parameters) for small β_t :

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t) = \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1}) q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))$$

Conditioned reverse process is a normal distribution with known parameters, describing how to denoise a noisy image $\mathbf{x}_t$ when we know the clean image $\mathbf{x}_0$.

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I})$$

Recap of Previous Lecture

- $\mathbf{z} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ represents the latent variables.
- Variational posterior distribution:

$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \dots, \mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}).$$

- Generative model and prior:

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = p_{\theta}(\mathbf{x}_0|\mathbf{x}_1); \quad p_{\theta}(\mathbf{z}) = \prod_{t=2}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \cdot p(\mathbf{x}_T)$$

Recap of Previous Lecture

Standard ELBO

$$\begin{aligned}\log p_{\theta}(\mathbf{x}) &\geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} = \mathcal{L}_{\phi, \theta}(\mathbf{x}) \rightarrow \max_{\phi, \theta} \\ \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) \\ &\quad - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}\end{aligned}$$

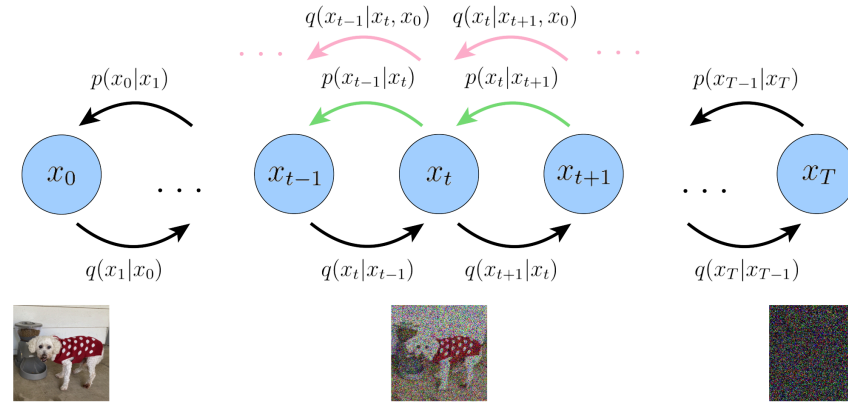
Outline

- 01 Diffusion ELBO Derivation (continued)
- 02 Gaussian Diffusion Reparametrization
- 03 Denoising Diffusion Probabilistic Model (DDPM)
- 04 Model Guidance
 - Classifier Guidance
 - Classifier-Free Guidance

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ELBO for Gaussian Diffusion Model



$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I}),$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))$$

ELBO for Gaussian Diffusion Model

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}), \\ p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))\end{aligned}$$

Let's assume that

$$\begin{aligned}\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t) &= \tilde{\beta}_t \mathbf{I} \\ \Rightarrow p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}).\end{aligned}$$

Theoretically, the optimal $\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t)$ lies in $[\tilde{\beta}_t, \beta_t]$:

- β_t is optimal for $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$;
- $\tilde{\beta}_t$ is optimal for $\mathbf{x}_0 \sim \delta(\mathbf{x}_0 - \mathbf{x}^*)$.

ELBO for Gaussian Diffusion Model

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}} \left(\mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \right. \\ &\quad \left. \parallel \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}) \right) \\ &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]\end{aligned}$$

ELBO for Gaussian Diffusion Model

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$.
3. Compute the ELBO:

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1 | \mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T)) \\ - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta, t}(\mathbf{x}_t)\|^2 \right]}_{\mathcal{L}_t}.$$

Sampling

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\theta, t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

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Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_0$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon} \quad \Rightarrow \quad \mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}}$$

- There is a linear relationship between $\boldsymbol{\epsilon}$, $\mathbf{x}_t$, and $\mathbf{x}_0$.
- Let's try to rewrite this mean using only $\mathbf{x}_t$ and $\boldsymbol{\epsilon}$.

Reparametrization of DDPM

$$\begin{aligned}\tilde{\mu}_t(\mathbf{x}_t, \epsilon) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t \\ &\quad + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \left(\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon}{\sqrt{\bar{\alpha}_t}} \right) \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} \cdot \epsilon\end{aligned}$$

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

Reparametrization

$$\begin{aligned} \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon} \\ \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t) \end{aligned}$$

Reparametrization of DDPM

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta, t}(\mathbf{x}_t) \right\|^2 \right] \\ &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta, t}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon) \right\|^2 \right]\end{aligned}$$

At every step of the reverse process, we attempt to predict the noise ϵ that was used in the forward diffusion process!

Reparametrization of DDPM

$$\begin{aligned}
 \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) \\
 &\quad - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t} \\
 \mathcal{L}_t &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta, t}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon) \right\|^2 \right]
 \end{aligned}$$

Reparametrization of DDPM

Let's drop the scaling coefficient; the reconstruction term ($t = 1$) has the same form.

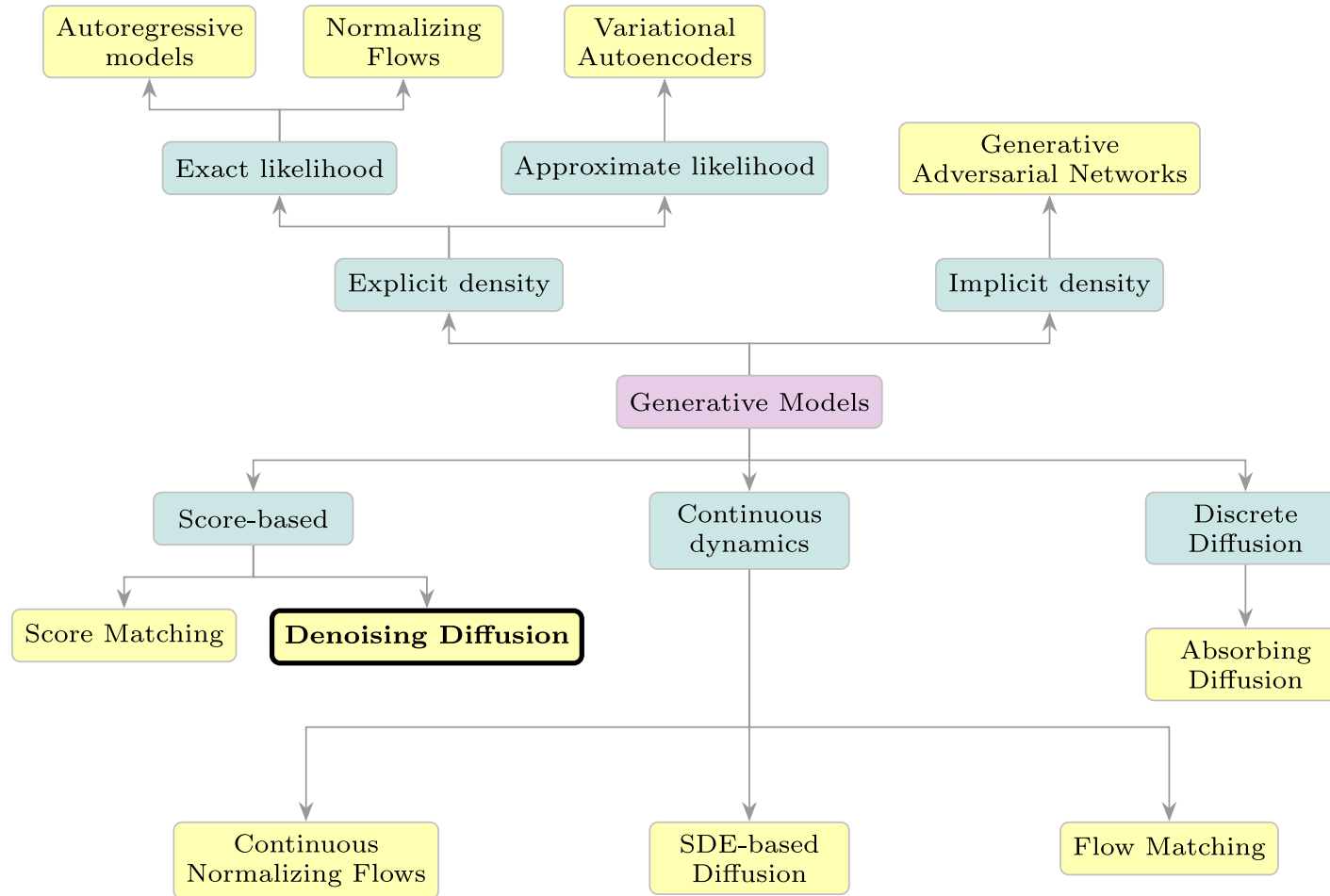
Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left\| \epsilon - \epsilon_{\theta, t} \left(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon \right) \right\|^2$$

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Generative Models Taxonomy



Denoising Diffusion Probabilistic Model (DDPM)

DDPM is a VAE Model

- The encoder is a fixed Gaussian Markov chain $q(\mathbf{x}_1, \dots, \mathbf{x}_T | \mathbf{x}_0)$.
- The latent variable is hierarchical (at each step, its dimension equals the input's).
- The decoder is a simple Gaussian model $p_\theta(\mathbf{x}_0 | \mathbf{x}_1)$.
- The prior distribution is given by a parametric Gaussian Markov chain $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$.

Forward Process

1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$;
2. $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t$;
3. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$.

Reverse Process

1. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$;
2. $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \cdot \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$;
3. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.

Denoising Diffusion Probabilistic Model (DDPM)

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$.
3. Compute loss $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$.

Sampling (Ancestral)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$:

$$\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t).$$

3. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

Denoising Diffusion as a Score-Based Generative Model

DDPM Objective

$$\begin{aligned}
 \mathcal{L}_t &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[C_{1,t} \cdot \left\| \epsilon_{\theta,t}(\mathbf{x}_t) - \epsilon \right\|_2^2 \right] \\
 &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[C_{2,t} \cdot \left\| \frac{\epsilon_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} - \frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}} \right\|_2^2 \right] \\
 q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbf{I}) \\
 \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) &= -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0}{1 - \bar{\alpha}_t} = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}}.
 \end{aligned}$$

Denoising Diffusion as a Score-Based Generative Model

We can reparameterize the model as:

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t) = -\frac{\boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t)}{\sqrt{1-\bar{\alpha}_t}} = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t).$$
$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[C_{2,t} \cdot \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) \right\|_2^2 \right]$$

DDPM vs NCSN: Objectives

DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[C_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

In practice, **this coefficient** is often omitted.

NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon$$

Maximizing the ELBO leads to the same objective as denoising score matching!

DDPM vs NCSN: Sampling

Sampling (Ancestral)

$$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}\mathbf{x}_{t-1} &= \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon} \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon} \\ &= \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}\end{aligned}$$

DDPM vs NCSN: Sampling

Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via L steps of Langevin dynamics at each noise level σ_t :

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \cdot \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \cdot \boldsymbol{\epsilon}_t^l.$$

3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to the next noise level σ_{t-1} .

DDPM vs NCSN: Summary

Summary

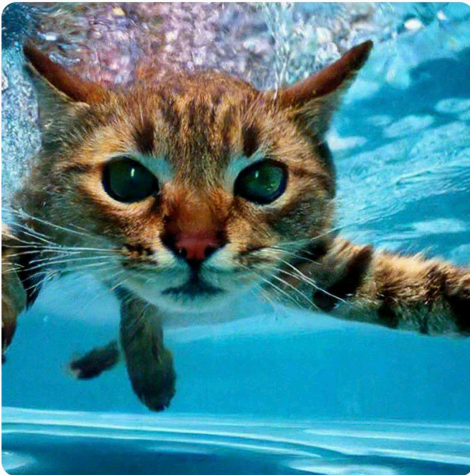
- Different Markov chains:
 - DDPM: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon;$
 - NCSN: $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon.$
 - One can generalize to $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$
- The objectives coincide: ELBO $\equiv$ score-matching.
- The sampling procedures differ:
 - Ancestral sampling in DDPM;
 - Annealed Langevin dynamics for NCSN;
 - Hybrid approaches that combine both updates are possible.

Outline

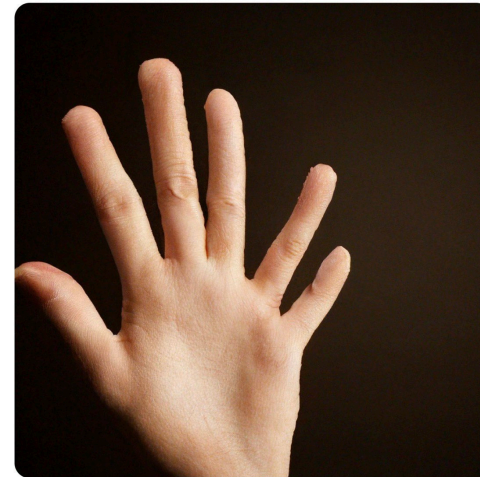
- 01 Diffusion ELBO Derivation (continued)
- 02 Gaussian Diffusion Reparametrization
- 03 Denoising Diffusion Probabilistic Model (DDPM)
- 04 Model Guidance**
 - Classifier Guidance
 - Classifier-Free Guidance

Guidance

- Up to now, we have focused on **unconditional** generative models $p_{\theta}(\mathbf{x})$.
- In practice, most generative models are **conditional** (in the diffusion era it is called guided): $p_{\theta}(\mathbf{x}|\mathbf{y})$.
- Here, $\mathbf{y}$ might denote a class label or **text** (as in text-to-image tasks).



Кот ныряет в бассейн, как ребенок на обложке альбома Nevermind, реалистично



рука человека с пятью пальцами, ни четырьмя, ни шестью, а с 5 (пять) пальцами

Conditional Models

In practice, we're typically interested in learning conditional models (sampling from conditional distribution $p_{\text{data}}(\mathbf{x}|\mathbf{y})$).

- $\mathbf{y} = \emptyset, \mathbf{x} = \text{image} \Rightarrow$ unconditional image model
- $\mathbf{y} = \text{class label}, \mathbf{x} = \text{image} \Rightarrow$ class-conditional image model
- $\mathbf{y} = \text{text prompt}, \mathbf{x} = \text{image} \Rightarrow$ text-to-image model
- $\mathbf{y} = \text{image}, \mathbf{x} = \text{image} \Rightarrow$ image-to-image model
- $\mathbf{y} = \text{image}, \mathbf{x} = \text{text} \Rightarrow$ image-to-text (image captioning) model
- $\mathbf{y} = \text{English text}, \mathbf{x} = \text{Russian text} \Rightarrow$ sequence-to-sequence model (machine translation)
- $\mathbf{y} = \text{sound}, \mathbf{x} = \text{text} \Rightarrow$ speech-to-text (automatic speech recognition) model
- $\mathbf{y} = \text{text}, \mathbf{x} = \text{sound} \Rightarrow$ text-to-speech model

Label Guidance

Label: Ostrich (10th ImageNet class)



VQ-VAE (Proposed)

BigGAN deep

Text Guidance

Prompt: a stained glass window of a panda eating bamboo

Left: $\gamma = 1$, Right: $\gamma = 3$.



Guidance in Generative Models

How to make a guided model?

Instead of sampling from $p_{\theta}(\mathbf{x})$, we sample from $p_{\theta}(\mathbf{x}|\mathbf{y})$.

Given **supervised** data $\{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$, we can treat $\mathbf{y}$ as an additional model input:

- $p_{\theta}(x_j|\mathbf{x}_{1:j-1}, \mathbf{y})$ for AR models;
- Encoder $q_{\phi}(\mathbf{z}|\mathbf{x}, \mathbf{y})$ and decoder $p_{\theta}(\mathbf{x}|\mathbf{z}, \mathbf{y})$ for VAEs;
- $G_{\theta}(\mathbf{z}, \mathbf{y})$ for NFs and GANs;
- $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$ for DDPMs.

Challenge

- Empirically, images sampled with this procedure do not fit well enough to the desired label $\mathbf{y}$.
- Being able to control the strength of guidance is especially valuable.

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Classifier Guidance

DDPM Sampling

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise the image (unconditional generation):

$$\begin{aligned}\mathbf{x}_{t-1} &= \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon \\ &= \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \sigma_t \cdot \epsilon\end{aligned}$$

Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) + \sigma_t \cdot \epsilon$$

What is the link between $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t)$ and $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y})$?

Classifier Guidance: Guided Score Function

Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) + \sigma_t \cdot \epsilon$$

Guided Generation

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) &= \nabla_{\mathbf{x}_t} \log \left(\frac{p_{\theta}(\mathbf{x}_t) p(\mathbf{y} | \mathbf{x}_t)}{p(\mathbf{y})} \right) \\ &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \\ &= \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \end{aligned}$$

Classifier Guidance: Guided Score Function

Guided Score Function

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}).$$
$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

Classifier Guidance: Guidance Scale

Guided Score Function

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

- Let us assume $\mathbf{y}$ is a class label.
- $p(\mathbf{y}|\mathbf{x}_t)$ is a classifier for noisy inputs.
- $p(\mathbf{y}|\mathbf{x}_t)$ is responsible for model guidance.

Guidance Scale

It is a natural idea to scale up the contribution of the guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

- The **guidance scale** γ adjusts the strength of classifier guidance.
- $\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y})$ is not the true guided score function $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.

Classifier Guidance: Distribution Sharpening

Scaled Guided Score Function

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

Scaled Conditional Distribution

$$\begin{aligned}\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\ &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)^{\gamma} \\ &= \nabla_{\mathbf{x}_t} \log \left(\frac{p_{\theta}(\mathbf{x}_t)p(\mathbf{y}|\mathbf{x}_t)^{\gamma}}{Z} \right)\end{aligned}$$

Note: Increasing γ sharpens $p(\mathbf{y}|\mathbf{x}_t)$, increasing the contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

Classifier Guidance: Overview

Training

1. Train the DDPM as before.
2. Train an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ on noisy data (time-dependent).

Sampling (Guided)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

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Classifier-Free Guidance

- The previous approach relies on training an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ for noisy images.
- We now introduce a method to sidestep this requirement.

$$\nabla_{\mathbf{x}_t}^\gamma \log p_\theta(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

Bayes theorem

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log \left(\frac{p_\theta(\mathbf{x}_t|\mathbf{y})p(\mathbf{y})}{p_\theta(\mathbf{x}_t)} \right) \\ &= \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) \end{aligned}$$

Classifier-Free Guidance

Scaled Guided Score Function

$$\begin{aligned}\nabla_{\mathbf{x}_t}^\gamma \log p_\theta(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\ &= \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + \gamma \cdot (\nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t)) \\ &= (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{y})\end{aligned}$$

Classifier-Free Guidance: Formulation

Scaled Guided Score Function

$$\begin{aligned}\nabla_{\mathbf{x}_t}^\gamma \log p_\theta(\mathbf{x}_t|\mathbf{y}) &= (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{y}) \\ \mathbf{s}_{\theta,t}^\gamma(\mathbf{x}_t, \mathbf{y}) &= (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})\end{aligned}$$

Naive training approach

- Train an unguided score function model $\mathbf{s}_{\theta,t}(\mathbf{x}_t)$.
- Train a guided score function model $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.
- Use their convex combination at inference.

Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^\gamma(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon$$

How to avoid training two separate score function models?

Classifier-Free Guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

CFG

1. Introduce the "absence of conditioning" label $\mathbf{y} = \emptyset$.
2. Identify the unguided score function $\mathbf{s}_{\theta,t}(\mathbf{x}_t) = \mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$.
3. Train a single model $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ on **supervised** data, dropping the label $\mathbf{y}$ with some fixed probability (simulating $\mathbf{y} = \emptyset$).
4. At inference, evaluate the model twice to obtain $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$ and $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.

Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

Summary

- At each step, DDPM predicts the noise that was injected in the forward process.
- DDPMs are quite slow, since the model must be applied T times for sampling.
- DDPM and NCSN are intimately connected at the objective level.
- Guidance technique allows to make controllable generation via conditioning on labels or text prompts.
- Classifier guidance turns an unconditional model into a conditional one by training an auxiliary classifier on noisy data.
- Classifier-free guidance removes the need for such a classifier, yielding a practical recipe now widely used.